Counterfactual Fluid Dynamics

Your run finished. Now answer the what-if — from the state the vehicle actually reached, rather than by rerunning each case from a checkpoint.

THE PROBLEM IT ADDRESSES

A case runs overnight. In the review, the first question is a what-if: three more degrees of bank, a hot-day atmosphere, ignition half a second later. Answering it today means a rerun per question, and the decision gets made on judgement before the results land. Interactions *between* domains are the ones that fall out between tools, because each tool solves its own domain and the results are composed afterwards.

WHAT IT DOES

It pauses a coupled run at a point the physics chooses, then continues every alternative from that one state, concurrently, into a single gated table with a per-branch record of why. Flow, chemistry, navigation and control run in one process over one shared field, so an interaction between them lands in the result. It sits above your solver the way a UQ or DoE driver does, on a different axis: branches from a state, not draws from an input distribution.

MEASURED, ON ONE DESKTOP MACHINE

Whole powered descent	**337.5 s**, 2516 coupled steps, four legs, sixteen gates
Peak memory	**18.3 MB** resident, five concurrent branches
Seventeen-branch campaign	**44.4 s** against a 600 s budget
Fan-out cost	**1.09×** one trunk step for five branches
Machine	Apple M3 Max, 16 cores, 128 GB

No cluster, no scheduler queue, no allocation request. Concurrent branches are bit-identical to the sequential run, held by a regression test.

VERIFIED AGAINST PUBLISHED REFERENCES

Manufactured solutions	observed order **1.98–2.01** against 2.00; divergence-free by construction
Taylor–Green (1937)	RHS kernel to **1.11e-16**, machine epsilon
Sod shock tube	density L1 **0.0175** against a 0.03 tolerance
RAM-C II flight, NASA (1970)	peak electron density **+0.35 decade** on the anchor

Thirteen self-verifying targets. Every one exits nonzero when its check fails, so a stale claim breaks a run rather than surviving in prose.

WHAT IT DOES NOT DO

- **It is not a replacement solver.** It reads no mesh format; geometry is declared in code on structured Cartesian grids. Meshing, converged flow at industrial scale, turbulence closures and certification stay with your existing solver.
- **Turbulence is staged, not available.** Validated incompressible cases sit at Re 100–1600.
- **One published result comes out with the wrong sign.** Supersonic retropropulsion drag collapse does not reproduce on this harness, and the cause is attributed to our own discretization rather than to the model class. It is documented with its numbers.
- **Distributed execution is a stated non-goal;** GPU work is deferred, with the reasoning published.

STANDING AND COST

MIT licensed, open source, published on crates.io. Part of the DeepCausality project, hosted at the Linux Foundation for Data & AI and maintained by the Center for Dynamic Causality. In continuous development since June 2023, across eight contributors. An engineer can clone it and have a gated result in under a minute on the machine already on their desk, without asking anyone's permission.